

Understanding Stability of Choices: Toward Robust Choice-Based Authentication in Cybersecurity

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Abstract—Traditional security questions (e.g., mother’s maiden name or first school) are easy to implement and use, but vulnerable to adversarial guessing or research, due to the increasing availability of personal information online, which reduces their effectiveness in cybersecurity authentication. Choice-based security questions offer a promising alternative by leveraging stable personal preferences that are harder for adversaries to predict. Prior studies have not comprehensively examined the temporal stability and consistency of users’ responses to a wide range of choice-based questions and their potential to improve the authentication. In this work, we study and analyze responses from 276 participants to nine choice-based questions (e.g., favorite color, number, season, day, animal, sport, month, food, and artist), collected three times with one-month intervals. Results show that questions on color, number, season, day, and animal exhibit high stability over a 2-month period, while maintaining high uncertainty against adversarial guessing, making them reliable candidates for robust authentication. In contrast, sport, month, food, and artist questions demonstrate lower stability. These findings highlight the importance of balancing stability (in terms of user memorability and recall consistency) with uncertainty (in terms of resistance in adversarial prediction) when designing choice-based questions, enabling more secure, user-friendly authentication mechanisms in cybersecurity systems.

Index Terms—Choice-based Authentication; Cybersecurity; Security Questions; Stability; Robust Authentication; Uncertainty; Usability.

I. INTRODUCTION

A. Motivation

The digital landscape is evolving rapidly, challenging traditional authentication mechanisms in cybersecurity systems that struggle to keep pace with vulnerabilities due to sophisticated cyber threats. As we depend more on digital solutions, the vulnerabilities in our cybersecurity systems become a more pressing issue to address. This rapid evolution necessitates innovative approaches to secure the authentication mechanism. Choice-based security questions represent a significant advancement in this context. By adapting to the changing landscape, such questions can leverage user-specific data less likely to be accessed by unauthorized entities, enhancing the overall cybersecurity systems [1].

Security questions based on personal facts, e.g., mother’s maiden name, city of birth, or first school, are widely used for password recovery and authentication due to their simplicity and user-centric design [2]. These security questions are increasingly examined for vulnerabilities in the evolving digital landscape [3]. Although traditional security questions

have long been a common mechanism in user authentication processes due to their simplicity and user-centric design, they are increasingly inadequate against sophisticated cyber threats. As highlighted in recent studies, including a Google report, the predictability of common security answers presents a significant risk; for instance, a hacker has a 39% chance of guessing users’ answers to the question “What is your city of birth?” [4].

Moreover, the increasing availability of personal information on social media and online platforms has significantly amplified risks, enabling social network mining, social engineering, and data breaches [5], [6]. Information previously regarded as sufficiently private for securing accounts, such as a user’s first pet’s name or mother’s maiden name, is now commonly available on social media platforms. This increased transparency enables social network mining attacks and allows acquaintances to readily guess answers, raising the risk of unauthorized access by individuals within a user’s social circle [7].

Considering the rapid evolution of cyber threats and the availability of personal information on online platforms, traditional security methods, e.g., passwords and fact-based security questions depending on easily accessible personal information, are increasingly becoming inadequate to secure user information. These approaches are vulnerable to social engineering and data breaches, as more personal information is exposed to digital platforms. This growing vulnerability underscores the urgent need in cybersecurity for robust mechanisms that utilize deeper and more personalized layers of a user’s identity, such as stable user preferences, which are significantly more challenging for adversaries to exploit [8].

Thereby, the need to enhance security questions’ reliability can be seen in critical settings, such as online banking and account recovery processes, where the strength of authentication measures can directly impact user security. Moreover, nearly 40% of users forget their security answers over time, complicating their usability and reliability. These limitations, including high predictability, exposure risk, and memorability issues (memory challenges), highlight the need for more robust alternatives in cybersecurity authentication, i.e., calls for reassessing security questions as a secure authentication method [9]. Therefore, there is a need to find a balance between user convenience and user-friendliness in terms of ease of remembering and stability, and the effective-

ness of security measures in terms of uncertainty to guess and hardness (resistance) to predict. Choice-based security questions, which leverage personal preferences (e.g., favorite color, number, season, day, or animal), present a promising solution. Unlike traditional fact-based questions, preferences are deeply inherent, less likely to be publicly documented or researched, and thus harder for adversaries to predict, while offering stability for consistent user recall. Unlike fact-based questions, preferences are deeply inherent, less likely to be publicly available or researched. This also makes it harder for adversaries to predict, while offering inherent stability for consistent user recall. This mechanism significantly reduces the burden to unlock the account and the associated support costs, offering a user-friendly alternative to traditional security questions.

Despite these advantages of choice-based questions, prior work has not comprehensively assessed the temporal stability and consistency of responses to a diverse set of choice-based questions, nor analyzed the potential of these questions to enhance authentication in cybersecurity systems through resistance to adversarial guessing. This gap impedes the design of reliable, user-friendly mechanisms that balance stability (for memorability and ease of use) with uncertainty (for resistance or hardness against prediction). Given the dual requirements for security questions: i) to be easy to remember for the user and ii) challenging enough for the adversaries to predict, this research seeks to uncover patterns that could inform the development of more robust cybersecurity questions that could be used to develop a robust passing sequence or password to mitigate cyber threats [10].

B. Related Work

Researchers have been investigating different factors and their effects on the efficiency of the authentication questions in cybersecurity systems. A study conducted by Ariel Rabkin found that the nature of security questions is scrutinized, particularly highlighting the shift in responsibility from users to system designers [11]. Rabkin emphasizes that the design of security questions plays a pivotal role in the efficacy of the authentication mechanism. Because the answer to a practical security question should naturally be stored in the user's long-term memory or be easily accessible, the user does not need to memorize it. This strategy aims to preserve security integrity while making the user authentication process simpler [11].

However, significant cybersecurity vulnerabilities are associated with security questions due to various efficient adversarial methods of procuring answers [12]. These techniques include automated data mining of personal details on social networks, as well as more intensive and focused human-driven efforts to gather data. Such methods are increasingly feasible and effective due to the vast amounts of personal information publicly available or easily obtained online [13]. As personal information becomes more available online, the security provided by traditional personal security questions is significantly compromised, rendering them less dependable as a secure form of user authentication.

These claims underline the need for re-evaluating security questions within authentication mechanisms in cybersecurity systems, suggesting a move towards a deeper understanding and secure methods that can adapt to the challenges posed by the digital age. Insights from the Nielsen et al. study deepen our understanding of how to improve security questions by leveraging recognition rather than relying solely on recall [14]. According to this research, authentication mechanisms rely on the cognitive processes of recognition and recall, which are crucial for user interaction with technology. Recognition-based methods, such as selecting a known item from a list, are highlighted for their user-friendly nature, easing the authentication process by providing cues that assist memory. In contrast, recall-based authentication requires users to retrieve information from memory without cues, which is a more challenging task and is less conducive to user convenience but better for security aspects. The objective is to improve the design of security questions, i.e., make robust and stable questions to make them easier for the user to respond to, while ensuring resistance (i.e., make it harder) for adversaries or unauthorized users to predict or get access to the system. Haga et al. [15] further developed the understanding of factors that help security questions in their foundational work, which evaluated question-and-answer passwords through empirical investigation. This study provided initial insights into the memorability and predictability of cognitive and associative passwords, highlighting their potential as robust password-creation methods.

Stainbrook et al. evaluated security and user convenience trade-offs, which sheds light on understanding how significant concerns impact user behavior and system design of memory-based authorization methods such as passwords and security questions [16]. According to the study, users often face a dilemma where they must choose between strong security measures that are cumbersome to use and more convenient options that might compromise security aspects. This trade-off is particularly evident in designing and selecting security questions for memory-based authentication mechanisms in cybersecurity systems. While it is easy to implement and use, traditional security questions can often be easily guessed or researched by adversaries, thus reducing their effectiveness as secure authentication measures. Findings presented by Stainbrook et al. suggest that authentication methods that incorporate personalization can help mitigate the traditional trade-offs between security and convenience. Thus, there is a need for a method to enhance security by utilizing information intimately linked to the individual and maintaining convenience, encouraging user compliance, and reducing the tendency to compromise on secure practices.

While choice-based security questions can be an option for authentication processes to deal with the current challenges that traditional memory-based security questions pose in today's digital age, choice-based security questions need to be assessed in terms of their stability over time in accordance with the stability guidelines mentioned by Julie Driebe and Tilmann Betsch [17], [18]. Betsch posits that once an attitude,

such as a preference, is formed and stored in memory, it lays the groundwork for stable, long-term recall that individuals can rely upon when re-encountering related decisions or questions. This ‘file-drawer’ concept of attitude storage means that individuals can efficiently retrieve their stored preferences when needed, making choice-based questions potentially more reliable for consistent authentication. However, Betsch also acknowledges that attitudes and preferences or choices are influenced by the situational context, suggesting that while preferences can offer a stable basis for security questions, their reliability may fluctuate depending on external factors. Nevertheless, preferences will likely guide behavior reliably in stable situations where environmental variables remain constant. Therefore, integrating choice-based security questions based on the stability of personal preferences/choices could enhance security in the authentication mechanism, making them a valuable addition to traditional cybersecurity systems [19]. Additionally, this can be a robust alternative to the complicated user authentication mechanisms explored in the literature [20], [21].

C. Contributions

This research examines the consistency of choice-based questions, focusing on their temporal stability and implications for cybersecurity systems. We conducted a study to develop a dataset where 276 individuals responded to a set of nine choice-based questions three times, with 1-month gaps between successive sets of responses. The study delves into the reliability and stability level of answers over time, employing graphical analyses and statistical tests, such as χ^2 - test and z - test, to evaluate the data. This study’s findings highlight the importance of choosing stable cybersecurity questions that are easy for users to remember and offering a significantly low challenge for them to answer, but at the same time, the answers should be hard to guess by adversaries; thereby contributing critical insights into enhancing the design of robust, secure, and user-friendly authentication mechanisms. The ultimate goal of this work is to find a strategy to develop a robust passing sequence or password using this set of choice-based questions and responses to these questions based on the relative reliability (in terms of difficulty to predict responses by adversaries) and stability (in terms of ease of consistency of responses and user-friendliness to remember over time). The passing sequence means a sequence of questions and correct responses to them, where the questions may appear in a random order. On the other hand, the password will have a fixed sequence of responses as a single word.

Organization: In this paper, we present our methodology, consisting of data collection and choice-based questions, data processing steps, and our analysis approach in Section II. After that, in Section III, we present our detailed graphical (both coarse-grained and fine-grained) and statistical analysis. Finally, we present our conclusion and discussion of our

takeaways along with the limitations and future direction in Section IV.

II. METHODS

This section outlines our approach to explore the stability of people’s responses to a set of choice-based questions asked three times with one-month gaps. We will first present our study and data collection details (Section II-A), followed by data processing steps (Section II-B). Finally, we will present our approach to analyze the data using graphical and statistical techniques (Section II-C).

A. Data Collection

In this study, 276 individuals responded to a set of nine choice-based questions (i.e., Q1 to Q9 in Figure 1), details are discussed below. Individuals responded to the same set of questions three times with a gap of one month (marked as set#1, set#2, and set#3 in the figure). Thereby, set#1 and set#3 are two months apart. These responses will be used to understand the stability of choices over time.

The nine choice-based questions are:

- **Question#1 (Q1)** – Choice of *Animal*: This choice-based question asks users their favorite *animal*, which could be used to construct a passing sequence or part of a robust password. Some of the commonly picked animals are gorillas, dogs, monkeys, and lions.
- **Question#2 (Q2)** – Choice of *Color*: This choice-based question asks users their *color* choice. This choice could be used to create a passing sequence or part of a robust password. Commonly picked colors are black, pink, or blue.
- **Question#3 (Q3)** – Choice of *Number*: This question asks users their favorite *number*, which could be used to produce a strong passing sequence or part of a robust password.
- **Question#4 (Q4)** – Choice of *Day*: This choice-based question asks users their preferred *day* of the week, like Friday or Saturday. While this could indicate personal routines or leisure activities tied to specific days, this *day*

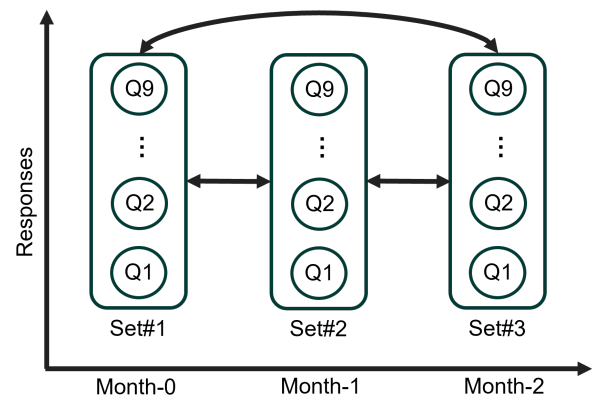


Fig. 1. Study timeline and response to a set of questions

choice could be used to create a robust passing sequence or part of a strong password.

- **Question#5** (Q5) – Choice of *Month*: This question asks users their favorite *month*. This choice could be used to make a powerful passing sequence or part of a strong password.
- **Question#6** (Q6) – Choice of *Season*: This choice-based question asks users their preferred *season*, which can relate to weather-related activities or personal comfort with climate conditions. This *season* choice could be used to create a robust passing sequence or part of a strong password.
- **Question#7** (Q7) – Choice of *Food*: This question asks users their favorite *food*, like pizza or tacos. The potential of using this choice as a passing sequence or password will be investigated in this work.
- **Question#8** (Q8) – Choice of *Artist*: This question asks users their favorite *artist*, such as Anirudh Ravichander or SZA. The potential of using this choice as a passing sequence or password will be investigated in this work.
- **Question#9** (Q9) – Choice of *Sport*: This question asks users their favorite *sport*, such as soccer or basketball. The potential of using this choice as a passing sequence or password will be investigated in this work.

B. Data Processing

The processing steps presented in this section address the need for uniformity and precision in handling survey responses. Some of the responses to Q1, i.e., choice of *animal*, fall into a super-set subset comparison. For instance, some individuals picked the dog in one set, and a dog breed, such as Husky, in another set. In this study, we resolve this super-set subset issue by converting the responses to super-sets, e.g., Husky to dog. Further, some people spell out a number, e.g., one, two, instead of the digits 1, 2. For uniformity, we convert users' responses to Q3, i.e., preferred *number*, to numeric values. While some people choose to spell out their favorite *month* partially or completely (e.g., Jan versus January), some prefer to use numeric scales, i.e., 1 – 12 to indicate January to December. For uniformity, we convert users' responses to Q5, i.e., choice of *month*, to the numeric scale. Next, all non-numeric responses are converted to lowercase and manually checked for human errors in responses, such as typing errors. Finally, for each question from the nine questions (Q1 – Q9), user responses between a pair of sets from the three sets, i.e., set#1 – set#3, are compared to determine the match count, i.e., stability of responses with varying time gaps. Based on these match counts, we define the following proportions or fractions.

- 1) p_1^k is obtained when comparing the responses of k^{th} question in set#1 and set#2, responded in 1-month gap
- 2) p_2^k is obtained when comparing the responses of k^{th} question in set#2 and set#3, responded in 1-month gap
- 3) p_3^k is obtained when comparing the responses of k^{th} question in set#1 and set#3, responded in 2-month gap

C. Analysis Approach

We will analyze the stability of people's responses using graphical and statistical techniques. While the bar graphs will demonstrate the overall stability (in terms of matching) in responses, the statistical tests will be used to demonstrate the significance of the findings.

1) *Graphical Approach*: We primarily use the bar graphs (with or without error bars) to compare the proportions of responses that match across the nine questions (Q1 – Q9) when they are responded to three times (i.e., sets#1 – 3).

2) *Statistical Approach*: We use the χ^2 – test and z – test to determine the significance of the findings from graphical analysis.

- 1) χ^2 – test is a statistical test we employed to determine whether the three proportions are significantly different. The null hypothesis for this test states that “there is no significant difference among the fractions of matches when the same question is responded to in 1-month and 2-month gaps.” That is, $H_0 : p_1^k = p_2^k = p_3^k$, where p_1^k , p_2^k , and p_3^k represent the proportions presented in Section II-B. At α -level of significance, if the p – value is below α , we conclude that we reject the null hypothesis, i.e., there is a significant difference among the three proportions. In other words, temporal differences have a significant impact on response matches, i.e., the stability of responses is affected by time. Otherwise, we fail to reject the null hypothesis, i.e., temporal differences do have a significant effect on response matching, i.e., stability. In summary, these tests will investigate the stability of users' choices across different questions answered with varying time gaps of one and two months.
- 2) z – test is a statistical test we employed to determine whether a pair of proportions out of three proportions is significantly different. The null hypothesis for this test states that “there is no significant difference among the fractions of matches when the same question is responded to in 1-month and/or 2-month gaps.” That is, $H_0 : p_i^k = p_j^k$, where p_i^k and p_j^k represent the two proportions presented in Section II-B with $i, j \in \{1, 2, 3\}$ and $i \neq j$. Similar to χ^2 – test, we use the α -level of significance when deciding whether we reject the null hypothesis. We use the z – test to compare a pair of proportions from the three proportions to determine which pair is significantly different when the χ^2 – test rejects the null hypothesis comparing the equality of three proportions for the same question responded three times.

Throughout this manuscript, p and $\hat{p}$ are used to indicate population and sample proportions. We will primarily use the $\alpha = 0.01$ significance level when comparing with the p – value to determine the statistical significance of a hypothesis test [22], [23]. Additionally, we use the $\alpha = 0.05$ level of significance.

III. ANALYSIS

In this section, we will analyze the temporal stability of users' responses to a set of nine choice-based questions (presented in Section II-A) answered over three months.

A. Graphical Analysis

In the graphical approach, we will first analyze the stability of responses to different choice-based questions in general using a coarse-grained analysis. After that, we will perform a fine-grained analysis of response stability across different time gaps, i.e., 1-month and 2-month gaps.

1) *Coarse-grained Analysis:* Figure 2 showcases the average stability of responses to various choice-based questions. This helps to show what choice-based questions have the lower and higher stability levels, when the same question is answered over time.

In the figure, we observe that the five choice-based questions, i.e., *color*, *season*, *number*, *animal*, and *sport* choice, demonstrate higher stability, i.e., average matched fraction of higher than 0.7, making them a good choice for creating a robust passing sequence or password. Additionally, responses to these questions have a wide range and are hard to predict by adversaries, compared to other choice-based questions, such as *month* and *day*, that have only a few options, i.e., 12 months or 7 days. In the figure, we observe a moderate stability of lower but close to 0.7 average matched fraction for the *month* and *day* choice questions. Therefore, these two choice-based questions are not as good as the top five questions with higher stability and reliability (due to the complexity of predicting responses by adversaries). Lastly, while the choice of *food* and *artist* might seem to have higher difficulty to predict by adversaries, they have a lower stability of average matched fraction below 0.6. So, these two questions are not a good choice for creating a recall-friendly passing sequence or password.

Thus, Figure 2 highlights the need for a balanced approach in selecting security questions – a mix that aligns stable choice

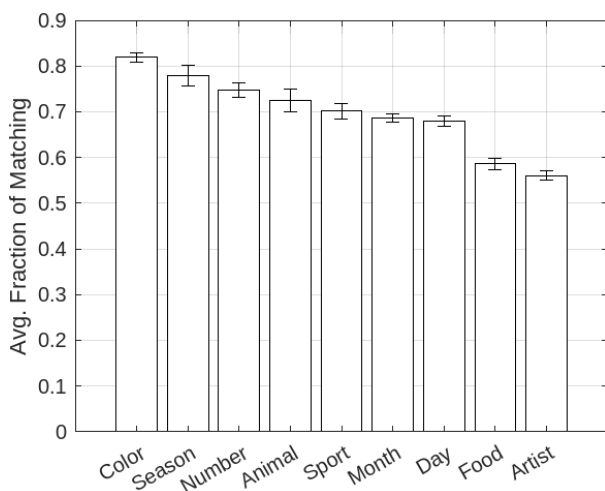


Fig. 2. Response stability over 1- and 2-month gaps

over time with a higher level of uncertainty for adversaries to predict. The ideal type of security questions should strike this delicate balance, maximizing the efficiency of the authentication process by being memorable for users to mitigate the burden of forgetting and resetting, as well as making it highly uncertain for adversaries to predict. The stability and uncertainty tradeoff enhances the overall user experience by reducing the likelihood of users forgetting answers and fortifying the security with higher uncertainty for adversaries to prevent breaches. A potential approach for constructing a robust but stable passing sequence or password could be to use more questions from the top five stable and highly uncertain questions (*color*, *season*, *number*, *animal*, and *sport*), and a few from the moderate/low stable and uncertain questions (*month*, *day*, *food*, and *artist*).

2) *Fine-grained Analysis:* This section examines the stability of responses to choice-based questions over different time gaps. The goal is to determine which questions yield the most stable response over time, i.e., could be a good choice to implement a resilient passing sequence or password.

Figure 3 (a) illustrates the fraction of matching across three pairs of sets, where each set comprises nine questions and is answered with a 1-month gap. Thereby, there is a 2-month gap between set#1 and set#3. The questions are sorted based on the average of the three fractions of matching (i.e., bars). In the figure, *color*, *season*, and *number* have an average fraction of matching above 0.7 across all three bars, suggesting that users' responses to these three questions are more stable over time, making these categories more stable or user-friendly for creating a resilient passing sequence or password. Also, due to a higher possibility of options, these three questions would also be hard for the adversaries to predict. So, they could be used to create a robust passing sequence or password. Four questions, i.e., *animal*, *sport*, *month*, and *day*, have moderate stability, as their average matched fraction is close to 0.7, with all of their 2-month gap average fraction of matches below 0.7 (third bars in Figure 3 (a)). So, they might need to be changed more frequently than the top three questions (*color*, *season*, and *number*) when used to make a passing sequence or password. Furthermore, for most of the questions, the 2-month gap average matched fraction is relatively lower than the 1-month gap average matched fraction (i.e., the first two bars in the figure). This pattern shows that the stability of responses is affected by time.

While in Figure 3 (b), five choice-based questions, i.e., *color*, *season*, *number*, *animal*, and *sport*, demonstrate higher stability, i.e., average matched fraction of higher than 0.7, with a gap of 1-month in responses, in Figure 3 (c), three choice-based questions, i.e., *color*, *season*, and *number*, demonstrate higher average matched fraction of higher than 0.7. In Figure 3 (b), two choice-based questions, i.e., *month* and *day*, and in Figure 3 (c), four choice-based questions, i.e., *animal*, *sport*, *month*, and *day*, demonstrate a moderate stability with an average matched fraction close to 0.7. So, they should be used in accordance with timely updates. Similar to Figure 2, the remaining two questions have low stability across both 1-

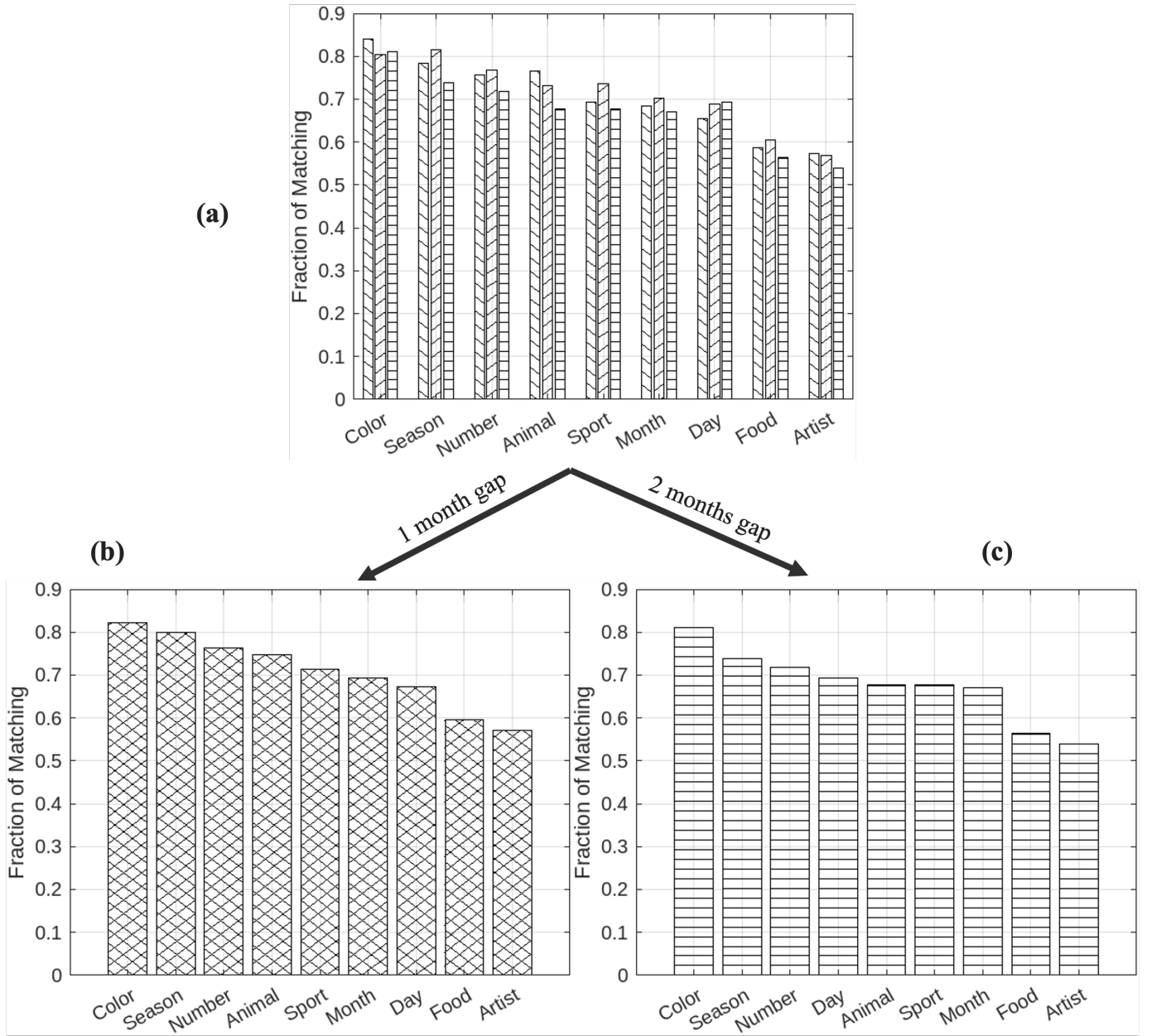


Fig. 3. Stability of users' responses to nine choice-based questions responded over time, with backslash ("\"), forward slash ("/), horizontal hatch ("---"), and cross hatch ("X") representing the response match between "set#1 and set#2" with 1-month gap, "set#2 and set#3" with 1-month gap, "set#1 and set#3" with 2-month gap, and combining the two 1-month gap response matches, respectively.

month and 2-month gaps, i.e., not a good choice due to ease of remembering.

B. Statistical Analysis

In this section, we perform the χ^2 proportion test (discussed in Section II-C) to determine the statistical significance of our findings from the previous section on graphical analysis. Table I summarizes the test findings. At the $\alpha = 0.01$ level of significance, we fail to reject the null hypothesis for all questions. Therefore, there is no significant difference among the proportions or fractions of matches when the same question is responded to in 1-month and 2-month gaps.

However, at the $\alpha = 0.05$ level of significance, we reject the null hypothesis for the *animal* and *season* questions. Therefore, there is a significant difference among the proportions or fractions of matches when those two questions are responded to in 1-month and 2-month gaps. For these two questions, we perform the z -test to determine which pairs of proportions are significantly different.

For the *animal* question, we find $z = 0.8825$ and $p = 0.37886$ ($\hat{p}_1$ versus $\hat{p}_2$), $z = 2.2776$ and $p = 0.0226$ ($\hat{p}_1$ versus $\hat{p}_3$), and $z = 1.3996$ and $p = 0.16152$ ($\hat{p}_2$ versus $\hat{p}_3$). Therefore, at the $\alpha = 0.05$ level of significance, we reject the

TABLE I
STATISTICAL COMPARISON OF RESPONSES ACROSS DIFFERENT SETS OF ITEMS (DF STANDS FOR DEGREES OF FREEDOM).

Item	$\hat{p}_1$	$\hat{p}_2$	$\hat{p}_3$	df	χ^2	p-value	$H_0 : \hat{p}_1 = \hat{p}_2 = \hat{p}_3$ at $\alpha = 0.01$
Animal	211/276	202/276	187/276	2	5.3384	0.0209	fail to reject
Color	232/276	222/276	224/276	2	1.3678	0.2422	fail to reject
Number	209/276	212/276	198/276	2	2.0865	0.1486	fail to reject
Day	181/276	190/276	191/276	2	1.0081	0.3154	fail to reject
Month	189/276	194/276	185/276	2	0.6841	0.4082	fail to reject
Season	216/276	225/276	204/276	2	4.6719	0.0307	fail to reject
Food	162/276	167/276	156/276	2	0.9059	0.3412	fail to reject
Artist	158/276	157/276	149/276	2	0.7158	0.3975	fail to reject
Sport	191/276	203/276	187/276	2	2.4002	0.1213	fail to reject

null hypothesis when comparing the $\hat{p}_1$ and $\hat{p}_3$. That is, there is a significant difference between the proportion or fraction of matches with 1-month and 2-month gaps between responses, and from Table I we witness that $\hat{p}_1 > \hat{p}_3$. Therefore, the proportion of matches with a 1-month gap is significantly higher than the proportion of matches with a 2-month gap. So, the choice of *animal* has relatively lower stability over time and will require a timely update.

For the *season* question, we find $z = -0.9557$ and $p = 0.33706$ ($\hat{p}_1$ versus $\hat{p}_2$), $z = 1.1974$ and $p = 0.23014$ ($\hat{p}_1$ versus $\hat{p}_3$), and $z = 2.1479$ and $p = 0.03156$ ($\hat{p}_2$ versus $\hat{p}_3$). Therefore, at the $\alpha = 0.05$ level of significance, we reject the null hypothesis when comparing the $\hat{p}_2$ and $\hat{p}_3$. That is, there is a significant difference between the proportion or fraction of matches with 1-month and 2-month gaps between responses, and from Table I we witness that $\hat{p}_2 > \hat{p}_3$. Therefore, the proportion of matches with a 1-month gap is significantly higher than the proportion of matches with a 2-month gap. So, the choice of *season* has relatively lower stability over time and will require a timely update. The stability of choice-based questions makes them excellent candidates for security questions because they will likely be remembered consistently over time, reducing the risk of incorrect answers that could lock users out of their accounts and create a burden to reset.

IV. CONCLUSION AND DISCUSSION

As stability (memorability and recall consistency) and uncertainty (hard for adversaries to predict) are essential for security questions, this study serves as a stepping stone toward enhancing the understanding that choice-based security questions can be the next step toward enhanced security in this authentication technique in cybersecurity systems. Furthermore, based on the temporal stability of choices across different questions, a strong passing sequence or robust password could be developed. This study analyzed responses to nine choice-based questions and their temporal stability (through graphical and statistical techniques), providing a comprehensive analysis that underscores their potential robustness in user authentication. Our empirical examination highlights the choice-based question categories best suited for robust authentication practices.

The graphical analysis illustrated that specific preference categories like *color*, *number*, *season*, *day*, and *animal* main-

tain high temporal stability. This consistency is not merely a statistical artifact but a reflection of how deeply these preferences are inherent in individuals' identities and daily lives. Unlike more transient security questions (hard to remember), such as the name of a favorite artist or food, which may change with passing trends or new experiences, fundamental preferences tend to be more stable and deeply rooted in personal life.

The statistical analysis reinforces these findings, with χ^2 - *test* revealing no significant difference over time for the most stable question categories at $\alpha = 0.01$. This lack of significant fluctuation over the 2 months shows the promise of the reliability of these categories in the authentication mechanism, where consistency is essential to security efficacy.

Choice-based questions offer a unique advantage over traditional information-based security queries (e.g., mother's maiden name, first school, and others). While the latter are often static and susceptible to being uncovered through social network mining, choice-based questions are dynamic and deeply personal. They are less likely to be guessed correctly by an adversary since they are inherently tied to an individual's tastes and experiences, which are not typically documented or shared.

Moreover, the stability of these preferences makes them particularly suitable for long-term memory retention without the cognitive burden of active recall, i.e., user-friendliness. Users are more likely to remember how they feel about fundamental aspects of their identity (like their favorite color or animal) than details of past events or data points, which often fade over time or become confused with other memories.

Incorporating choice-based questions to create a passing sequence or password with questions/responses from the 3-5 questions that have higher stability and more options for responses, i.e., higher uncertainty or resistance for the adversaries to predict, could help to develop an authentication mechanism that can enhance both security and user-friendliness in the following aspects:

- **Enhanced Security:** By leveraging the natural consistency in personal preferences, organizations can reduce the risk of unauthorized access due to predictable answers. The low predictability or high uncertainty of deep-seated preferences compared to factual information makes them

more challenging for adversaries to crack.

- Improved User Experience: Users benefit from intuitive and memorable security questions that reduce the frustration and potential lockouts associated with forgetting the answers. This can improve the user-friendliness and compliance with security protocols and decrease the need for costly and insecure password resets.

While choice-based security questions offer considerable advantages over traditional fact-based questions, this work has some limitations. One limitation arises from the cultural and demographic differences that can influence preference stability. Preferences such as favorite color, animal, or sport may be less uniformly enduring across different cultures or age groups. For example, younger individuals might change their preferences more frequently as they are exposed to new experiences and information. Similarly, cultural factors can play a crucial role in shaping and reshaping preferences over time, which could affect the overall reliability of these questions in diverse user populations.

Although personal preferences are generally stable, they remain susceptible to change due to significant life events, evolving interests, or shifts in personal and social circumstances. The reliability of preference-based security questions in this study depends on the assumption of consistency over a two-month period. However, this may not persist over longer durations. Future research should therefore conduct larger-scale, longitudinal studies to better understand the conditions under which preference stability diminishes and when periodic resets of authentication choices become necessary. In addition, there is a delicate balance between choosing memorable and secure questions. Questions that are too unique may be secure but risk being forgotten, while very common preferences may be easy to remember but also easy to guess or research by adversaries. Finding the right questions that strike this balance is challenging and requires careful consideration.

While choice-based security questions offer a promising alternative to traditional security questions by leveraging deep-seated personal preferences, these limitations highlight the need for a balanced approach. By carefully selecting culturally and demographically appropriate questions, balancing memorability with security, and continuously evaluating their effectiveness and acceptance among users, organizations can effectively incorporate these questions into their authentication mechanisms to secure cyber systems. Such an approach enhances security and respects the complexity of human preferences and the dynamics of memory, thereby contributing to more robust and user-friendly authentication mechanisms.

ACKNOWLEDGMENT

We would like to thank the study participants for their contribution to this research.

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